Linear approximation. for $x-a \to 0$

$$f(x) = f(a) + f'(a)(x-a)$$

Quadratic approximation

$$f(x) = f(a) + f'(a)(x-a) + \frac{1}{2} f''(a)(x-a)^2$$

for $x \to 0$

$f(x)$	Linear	Quadratic
$\sin x$	$\approx x$	$\approx x$
$\cos x$	≈ 1	$\approx 1 - \frac{x^2}{2}$
e^x	$\approx 1+x$	$\approx 1+x+\frac{x^2}{2}$
$\ln(1+x)$	$\approx x$	$\approx x - \frac{x^2}{2}$
$(1+x)^r$	$\approx 1+rx$	$\approx 1+rx+\frac{r(r-1)}{2}x^2$

$$a_k = (1+\tfrac{1}{k})^k$$

$$\ln a_k = k \cdot \ln(1+\tfrac{1}{k}) \approx k \cdot \tfrac{1}{k} = 1$$

$$k \to \infty \quad \tfrac{1}{k} \to \infty$$

$$\ln a_k = 1 \qquad a_k = e$$

$(w(x)+1)\cdot e^{w(x)} = x$. (Hint, $w(1)=0$)

use linear approximation to estimate $w(\frac{3}{2})$

$$w(\tfrac{3}{2}) = w(1) + w'(1)\cdot(\tfrac{3}{2}-1) = \frac{w'(1)}{2}$$

$$\frac{dw}{dx}\cdot e^{w(x)} + \frac{dw}{dx}(e^{w(x)}\cdot(w(x)+1)) = 1.$$

$$\frac{dw}{dx} = \frac{1}{e^{w(x)} + e^{w(x)}\cdot(w(x)+1)}$$

$$w'(1) = \frac{1}{e^0 + e^0\cdot 1} = \frac{1}{2}$$

$$w(\tfrac{3}{2}) = \frac{1}{2}w'(1) = \frac{1}{4}$$

Product of Linear approximation

$$(f\cdot g)(x) = f(a)g(a) + (f'(a)g(a) + g'(a)f(a))\cdot(x-a)$$

$x \to a$

To show $Q(fg) = Q(Q(f)\,Q(g))$

cheat sheet.

$(fg)' = f'g + fg'$

$(fg)'' = f''g + g''f + f'g'$

right side:

$$Q(f) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2$$

$$Q(g) = g(0) + g'(0)x + \frac{g''(0)}{2}x^2$$

$$Q(Q(f)\cdot Q(g)) = (fg(0) + (f'g(0) + fg'(0))x + \left(\frac{fg''(0)}{2} + \frac{gf''(0)}{2} + f'g'(0)\right)x^2 + (\cdots)x^3 + (\cdots)x^4)$$

(drop higher-order terms)

left side:

$$Q(fg) = fg(0) + (fg)'(0)x + \frac{(fg)''(0)}{2}x^2$$

$$= fg(0) + (f'g(0) + fg'(0))x + \left(\frac{fg''(0)}{2} + \frac{gf''(0)}{2} + f'g'(0)\right)x^2$$